UNIVERSITY OF LONDON

CM2025

BSc EXAMINATION

COMPUTER SCIENCE

Computer Security

Release date: Thursday 15 September 2022 at 12:00 midday British Summer Time

Submission date: Friday 16 September 2022 by 12:00 midday British Summer Time

Time allowed: 24 hours to submit

INSTRUCTIONS TO CANDIDATES:

Section A of this assessment paper consists of a set of **TEN** Multiple Choice Questions (MCQs) which you will take separately from this paper. You should attempt to answer **ALL** the questions in Section A. The maximum mark for Section A is **40**.

Section A will be completed online on the VLE. You may choose to access the MCQs at any time following the release of the paper, but once you have accessed the MCQs you must submit your answers before the deadline or within **4 hours** of starting whichever occurs first.

Section B of this assessment paper is an online assessment to be completed within the same 24-hour window as Section A. We anticipate that approximately **1 hour** is sufficient for you to answer Section B. Candidates must answer **TWO** out of the THREE questions in Section B. The maximum mark for Section B is **60**.

You may use any calculator for any appropriate calculations, but you may not use computer software to obtain solutions. Credit will only be given if all workings are shown.

You should complete **Section B** of this paper and submit your answers as **one document**, if possible, in Microsoft Word or a PDF to the appropriate area on the VLE. Each file uploaded must be accompanied by a coversheet containing your **candidate number**. In addition, your answers must have your candidate number written clearly at the top of the page before you upload your work. Do not write your name anywhere in your answers.

© University of London 2022
UL22/1060

SECTION A

Candidates should answer the **TEN** Multiple Choice Questions (MCQs) quiz, **Question 1** in Section A on the VLE.

SECTION B

Candidates should answer any **TWO** questions from Section B.

Question 2

Computer security and ethics

At the beginning of the Computer Security course, you were instructed to familiarise yourself with the following article:

”Ethics in Information Security” in IEEE Security & Privacy, vol. 15, no. 03, pp. 3-4, 2017.

Imagine a scenario in which a government agency learns of a security flaw in a smartphone operating system. The security flaw allows an attacker to remotely access the device and to do and see anything the owner of that device could do and see. The flaw includes the ability to unlock the device.

(a) Based on the discussion in that article, make ONE argument for the government agency to disclose the vulnerability and ONE argument against the government agency disclosing the vulnerability to the smartphone manufacturer. [6]

(b) Explain what ’timely, responsible disclosure’ is in terms of security vulnerabilities. [2]

(c) Why is ’timely, responsible disclosure’ considered best practice in the security industry with regard to security vulnerabilities? [3]

(d) The article presents the example of a document collection released by WikiLeaks called ’Vault 7’. What was Vault 7? [2]

(e) Investigate ’Vault 7’ on the Internet. Do you think ’Vault 7’ was a timely and responsible disclosure? Make TWO points for and TWO points against this point of view. [8]

(f) In the 2018 ACM code of ethics available on https://ethics.acm.org/, there are 3 top-level sections:

i. GENERAL ETHICAL PRINCIPLES.

ii. PROFESSIONAL RESPONSIBILITIES.

iii. PROFESSIONAL LEADERSHIP PRINCIPLES.

You are working on a web application which will store private data such as names, addresses and phone numbers of the users. You realise that it has a security vulnerability which would allow a normal user to gain access to the private details of other users, which they should not be able to do. Name THREE subsections of the ACM code of ethics which give reasons for fixing this vulnerability, even though your manager tells you there is not time.

For each subsection you choose, in your own words explain why that subsection relates to the problem. [9]

Question 3

Cryptography

(a) In your own words, explain how the simple Caesar shift cipher works. Make it clear what the inputs, parameters and outputs are. Use an example if it helps your explanation. [4]

(b) Explain why decoding a Caeser-encoded string is harder than encoding, assuming you do not know the shift when decoding. [2]

(c) Explain how a computer program that can decode a Ceaser-encoded string automatically could work. [2]

(d) In the grid-based transposition cipher seen in the course, a string is encrypted as follows:

i. Select a number of columns C.

ii. Write the input string into a grid with C columns and enough rows to contain the whole string. Write from left to right, returning to the leftmost column when you reach the end of a row.

iii. Fill empty cells at the end with X.

iv. To obtain the encrypted string, read from top to bottom of each column, moving back to the top and across to the next column when you have read the bottom row.

Here is an example of encoding the string 'IAMTHESPY':

Write to a grid with $C = 3$:

I	A	M
T	H	E
S	P	Y

Read off column-wise: "ITSAHPMEY"

Encode the first 12 characters of your name using this method, with a C of 4. Present the input string (your name), show your grid and write the output string. [3]

(e) A more advanced version of this cipher uses a key word to re-order the columns prior to reading off the encrypted text.

Write the key word at the top:

B	A	D
I	A	M
T	H	E
S	P	Y

Re-order the columns by sorting the letters in the key word alphabetically:

A	B	D
A	I	M
H	T	E
P	S	Y

Read off the letters, ignoring the key word row: ”AHPITSMEY”

Encode the first 12 letters of your name with key word CLEVER (and therefore, $C = 6$). Show your grid before and after re-ordering. Also write the string before and after it is encoded. [4]

(f) Write a working, text based program that can encode using the key word version of the transposition cipher you have just been using. The program should allow for varying sizes of key word. Your answer to this question should consist of the code of your program and the text output of your program when it is encoding the phrase ’the packet is in the letterbox’ using key words that are the first three, four and five letters of your name. Do not paste a screenshot, copy paste the code and text output of the program. Use any programming language you like.

[15]

Question 4

RSA and Blockchain

(a) What does parity checking mean for binary data? [2]

(b) Explain how you could use hash functions to check if some data has been received intact. [4]

(c) You are working on a system which stores passwords in a database. Would it be secure to store plaintext passwords in the database? Assume access to the database is controlled via passwords. Justify your answer. [2]

(d) How could you use one-way hashing to store the passwords securely and how would you then check passwords for users who try to login using the accounts stored in the database? [6]

(e) In the RSA cryptosystem, users publish their public keys. Drawing on your understanding of how RSA works, explain why it is secure for people to publish their public keys. [6]

(f) List THREE things that you find in every Bitcoin block and state their purpose. [6]

(g) Explain how the Bitcoin design prevents a bad actor from adjusting the contents of blocks in the chain. [4]

END OF PAPER